

NAGARAJU SANDRALA

Applied AI Engineer · Forward Deployed Engineer (FDE)

LLMs · RAG Pipelines · AI Agents · Python · Claude & Anthropic Ecosystem · AWS Bedrock · LangGraph · MLOps

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PROFESSIONAL SUMMARY

Applied AI Engineer and Forward Deployed Engineer with 14+ years of production experience building LLM-integrated applications, end-to-end RAG pipelines, multi-step AI agent systems, and enterprise data infrastructure in regulated public health, transportation, and government environments.

Deep expertise across the complete AI engineering stack: Claude and Anthropic ecosystem (Messages API, tool use, MCP), LangGraph StateGraph agents with fault tolerance and human-in-the-loop, RAG pipeline design with RAGAS-based evaluation, HIPAA-compliant AI deployment on AWS Bedrock and SageMaker, MLOps with CI/CD evaluation gates, and PowerShell DSC infrastructure-as-code for HA enterprise systems.

Uniquely experienced navigating the full lifecycle from customer problem → rapid technical solution → institutional AI governance approval → production deployment: including the first locally hosted LLM application approved for production use in a South Carolina state agency after a four-month AI committee process.

PRODUCTION AI IMPACT

1.9M+ mosquito-borne cases tracked (7-app AI surveillance platform) | **46** South Carolina counties served by production LLM (first in state agency) | **<4 hrs** full HA ArcGIS Enterprise from bare OS via PowerShell DSC | **14+ yrs** production AI/ML and enterprise engineering experience

TECHNICAL SKILLS

Claude & Anthropic

Claude Sonnet 4.5 / Haiku / Opus — Anthropic API and AWS Bedrock; Messages API (tool use, streaming, vision, multi-turn); system prompt grounding and instruction hierarchy; 200K context window management; Anthropic MCP (server/client, dynamic tool discovery, tool allowlisting); Claude Code agentic workflows; prompt injection defense

LLMs & GenAI

Llama 3 (Ollama, on-prem), GPT-4o (OpenAI API), AWS Bedrock (Titan, Nova, Claude), Amazon SageMaker; multi-model routing with LiteLLM; provider fallback and circuit breaker; prompt engineering; structured JSON output; temperature=0 for faithfulness; HyDE; RAG-Fusion

RAG Pipelines

Embedding model selection and domain benchmarking (BGE-M3, Titan, ada-002, MiniLM), semantic/hierarchical/parent-child chunking, hybrid BM25+dense retrieval with RRF, Cohere Rerank cross-encoder, MMR diversity, top-k optimization, lost-in-the-middle mitigation

Vector Databases

Pinecone (namespace sharding, HNSW tuning), FAISS (IVF+PQ, on-prem), OpenSearch k-NN (sub-second ANN at scale), pgvector on RDS, Redis cache layer

Agents & Orchestration

LangGraph (typed StateGraph, SqliteSaver checkpointing, human-in-the-loop interrupt, fan-out/fan-in via Send API, conditional edges), LangChain/LCEL, ReAct, Plan-and-Execute, supervisor-based multi-agent routing, MCP enterprise integrations

Evaluation & MLOps

RAGAS (faithfulness, context recall/precision, answer relevancy), LLM-as-judge with anti-bias protocol, G-Eval, LangSmith tracing, MLflow experiment tracking and model registry, Databricks Asset Bundles, CI/CD evaluation gates

Cloud & Infrastructure

AWS Bedrock (VPC endpoints, HIPAA-eligible, BAA), Amazon SageMaker, AWS ECS Fargate (SQS autoscaling), DynamoDB, GitHub Actions, Azure DevOps CI/CD, Docker, PowerShell DSC IaC, Secrets Manager, CloudWatch

Guardrails & Safety

PII/PHI detection (AWS Comprehend Medical, Presidio), prompt injection detection, confidence-based routing, faithfulness guards, HIPAA audit logging, VPC data residency, async human-in-the-loop escalation

Languages & APIs

Python 3.x, C#/I.NET, JavaScript/TypeScript, PowerShell, T-SQL, Flask, FastAPI, ASP.NET Core Web API, ArcGIS REST API, Tableau JWT OAuth2

Data & Databases

SQL Server, PostgreSQL/PostGIS, DynamoDB, Redis, SQS, SDE geodatabases, large-scale ETL (10M+ documents), async batched processing, schema-validated agent outputs

PROFESSIONAL EXPERIENCE

Applied AI Engineer · Forward Deployed Engineer · GIS Enterprise Architect

South Carolina Department of Public Health (SCDPH) — Remote — Dec 2021 – Present

- Designed and deployed the first locally-hosted LLM application approved for production in a South Carolina state agency: full-stack AI chatbot for mosquito-borne disease surveillance using Ollama (Llama 3), Flask REST API, and ArcGIS Maps SDK for JavaScript — connected live to ArcGIS Online FeatureServer endpoints across all 46 South Carolina counties. Zero PHI leaves the government network.
- Navigated the complete SCDPH AI governance approval process (AI Use Case Impact Assessment, Third-Party Software Questionnaire, State IT Plan Form) — a four-month institutional compliance cycle, the first LLM approval of its kind at the agency.
- Designed multi-agent workflows using LangGraph orchestration with supervisor-based routing, coordinating specialized agents for classification, rule retrieval, and remediation across governance and support use cases.
- Built hybrid RAG pipelines combining semantic and keyword search over vector-indexed knowledge bases using AWS Bedrock embedding models, grounding agent responses in governance rules and domain documentation.
- Deployed Claude Sonnet and Haiku via AWS Bedrock and Anthropic API with tool use, streaming, multi-turn memory, and 200K context window for multi-document synthesis.
- Implemented Anthropic MCP server/client architecture for enterprise tool integration, connecting agents to ArcGIS REST API, SQL Server, and internal data services without hardcoded function schemas.
- Engineered mosquito-borne clinical AI surveillance infrastructure spanning seven AI applications sustaining 30-40× baseline traffic without downtime across 5+ years of continuous production operation.
- Architected and deployed HA ArcGIS Enterprise using PowerShell DSC IaC — complete environment from bare OS in under 4 hours, zero-downtime rolling updates with automated health verification.

AI-Enabled Platform Engineer · Forward Deployed Engineer · GIS Specialist

Texas Department of Transportation (TxDOT) — Remote (concurrent engagement) — Oct 2021 – Oct 2025

- Designed agentic AI workflows using LangChain and LangGraph — multi-step reasoning and tool-calling pipelines integrating LLMs with internal data sources via RAG and MCP. Built production MCP servers enabling AI coding agents (Claude, Copilot, Cursor) to connect to enterprise tools and data sources.
- Built embedding-based semantic search with chunking/indexing strategies optimized for retrieval accuracy; designed and benchmarked vector DB infrastructure (OpenSearch k-NN, Pinecone, FAISS) supporting sub-second ANN retrieval across millions of document chunks.
- Led development of an enterprise MLOps platform integrating Databricks, MLflow, and AWS services for standardized model development, deployment, monitoring, and governance across DEV/STAGING/PROD.
- Designed and deployed the TxDOT Nighttime Inspection Suite — a 3-application mobile AI platform deployed statewide across ~80,000 centerline miles of US highway for FHWA-mandated compliance.
- Mentored engineering teams on MLOps architecture, CI/CD automation, and scalable ML platform design.

GIS Developer / Admin · Decision-Support Systems

Florida Department of Transportation (FDOT & DMS) — Lake City, FL — Mar 2017 – Oct 2021

- Supported hurricane evacuation network GIS decision-support systems during active hurricane events, providing real-time geospatial intelligence to emergency managers.
- Designed ASP.NET Core Web API services, ArcGIS JavaScript API applications, and enterprise geodatabase pipelines for statewide transportation asset management; maintained 110,000+ parcel geodatabases.

GIS Developer / Analyst

AT&T — Alpharetta, GA — Oct 2016 – Feb 2017

- Developed GIS web applications using ArcGIS API for JavaScript and Silverlight; integrated external datasets into Roads and Highways for HPMS and Certified Public Mileage reporting; migrated legacy ASP/VB.NET systems to ASP.NET Core.

GIS Developer / Analyst

L Brands · Value Momentum Pvt Ltd — Multiple Locations — Jul 2010 – Sep 2016

- GIS web application development, ArcFM utility network tools (C#, VB6), enterprise geodatabase design and SDE migration, SQL Server database design, RESTful API integrations across utility, insurance, and government verticals.

EDUCATION

Bachelor of Science in Computer Science — Osmania University, Hyderabad, India